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Modelling Crypto–Currencies Financial Time–Series

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Abstract

This paper studies the behaviour of crypto–currencies financial time–series of which Bitcoin is the most prominent example. The dynamic of those series is quite complex displaying extreme observations, asymmetries, and several nonlinear characteristics which are difficult to model. We develop a new dynamic model able to account for long–memory and asymmetries in the volatility process as well as for the presence of time–varying skewness and kurtosis. The empirical application, carried out on a large set of crypto–currencies, shows evidence of long memory and leverage effect that has a substantial contribution to the volatility dynamic. Going forward, as this new and unexplored market will develop, our results will be important for asset allocation, risk management, and pricing of derivative securities.

Keywords: Crypto-currency; Bitcoin, Score–Driven model; Leverage effect; Long memory; Higher Order Moments

1. Introduction

Bitcoin is the first decentralized cryptocurrency created in 2009 and documented in Nakamoto (2009). Since its inception, it gained a growing attention from the media, academics, and finance industry, and in recent months the global interest in Bitcoin and crypto–currencies has spiked dramatically. The reasons for this are several, just to name few: Japan and South Korea have recognized Bitcoin as a legal method of payment (Bloomberg, 2017a; Cointelegraph, 2017); some central banks are exploring the use of the crypto–currencies (Bloomberg, 2017c); a large number of companies and

banks created the Enterprise Ethereum Alliance[1] to make use of the crypto–currencies and the related technology called blockchain, (Forbes, 2017). Finally the Chicago Mercantile Exchange (CME) will start to negotiate Bitcoin futures on 18th of December 2017, see Exchange (2017a), Nasdaq and Tokyo Financial Exchange will follow in 2018, see (Bloomberg, 2017b; Exchange, 2017b).

All this interest has been reflected on the crypto–currencies market capitalization that exploded from around 19 billion in February 2017 to around 355 billion in December 2017.[2] For this reason, we believe it is time to analyze in depth this new market that is becoming an important piece of financial technology.

Although Bitcoin can be considered to be relatively new, there has already been some initial analysis into the crypto–currency. Hencic and Gourieroux (2014) applied a non–causal autoregressive model to detect the presence of bubbles in the Bitcoin/USD exchange rate. Sapuric and Kokkinaki (2014) measure volatility of Bitcoin exchange rate against six major currencies. Chu et al. (2015) provide a statistical analysis of the log–returns of the exchange rate of Bitcoin versus the USD. They found that the Generalized Hyperbolic distribution seems to be the most appropriate choice to model the unconditional distribution of crypto–currencies time–series.

The contribution of the paper is twofold. First, we develop a dynamic updating equation for the volatility of a time–series exploiting the information contained in the score of the conditional distribution of the data, which is assumed to follow the Generalized Hyperbolic Skew Student's t law (GHSKT); this distribution is detailed in Aas and Haff (2006) and obtained as a limiting case of the Generalized Hyperbolic family (Barndorff-Nielsen, 1977). This dynamic model can be thought of as a special case of the multivariate model introduced by Lucas et al. (2014) and Lucas et al. (2017) and belongs to the Score–Driven class of models recently introduced by Creal et al. (2013) and Harvey (2013). Our contribution extends that of Lucas et al. (2014) and Lucas et al. (2017) in several directions, namely we: i) derive expressions in order to apply the model in a univariate framework, these cannot be easily recovered from Lucas et al. (2014), ii) investigate in deep the characteristics of

[1]Source: https://entethalliance.org/members/

[2]Source: https://coinmarketcap.com/

the filter implied for the volatility update, especially the implications of the asymmetric tail behaviour of the GHSKT distribution, and iii) extend the model in order to account for several features such as the presence of leverage effects (Black, 1976); long memory properties of the volatility process and time–varying higher order moments. Furthermore, we report a Monte Carlo experiment in order to investigate the finite sample properties of the Maximum Likelihood estimator.

The second contribution of this paper is empirical. Specifically, we concentrate on the dynamic properties of crypto–currencies financial time–series and not on their unconditional distribution as in previous studies, see Chu et al. (2015). Moreover, we investigate a large set of crypto–currencies, which to the best of our knowledge has not been done yet.

The paper proceeds as follows: Section 2 presents the basic model. Section 3 reports the Monte Carlo experiment. Section 4 reports the extensions of the model in order to incorporate: leverage effects, long memory into the volatility process, and time–varying higher order moments. Section 5 provides the main empirical contributions of the paper. Section 6 draws some conclusions.

2. The Model

Let y_t be the realization of the random variable $Y_t \in \Re$ at time t, we assume:

$$Y_t = \varsigma + \exp(h_t)\varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} \mathcal{GHSKT}(0, 1, \boldsymbol{\theta}), \tag{1}$$

$$h_t = \phi(h_{t-1}, y_{t-1}; \boldsymbol{\phi}), \tag{2}$$

where ς is a constant, h_t is the log–volatility obtained as a function of the past log–volatility level, h_{t-1}, and the realization y_{t-1}, via the known function $\phi : \Re \times \Re \to \Re^+$. The function ϕ constitutes the filter for the volatility and characterises the model specification, for example, when $\phi(h_{t-1}, y_{t-1}, \boldsymbol{\psi}) = (1/2)\log(\phi_0 + \phi_1(y_{t-1} - \varsigma)^2 + \phi_2 \exp(2h_{t-1}))$, where $\boldsymbol{\psi} = (\phi_0, \phi_1, \phi_2, \varsigma)'$, we obtain the popular GARCH specification of Bollerslev (1986). To conclude the model specification, the only source of error affecting the dynamic system, ε_t, is assumed to be distributed independently as a standardized GHSKT. We briefly detail how to construct this distribution in the next subsection, for the sake of notation we drop the time index t.

2.1. The distributional assumption

The GHSKT distribution detailed in Aas and Haff (2006) has probability density function (PDF):

$$p(y;\boldsymbol{\theta}^*) = \frac{2^{(1-\nu)/2}\delta^{\nu}|\beta|^{(\nu+1)/2}K_{(\nu+1)/2}\left(\sqrt{\beta^2(\delta^2+(y-\mu)^2)}\right)\exp(\beta(y-\mu))}{\Gamma(\nu/2)\sqrt{\pi}(\sqrt{\delta^2+(y-\mu)^2})^{(\nu+1)/2}}, \quad \beta \neq 0, \tag{3}$$

and

$$p(y;\boldsymbol{\theta}^*) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi}\delta\Gamma(\frac{\nu}{2})}\left[1+\frac{(y-\mu)^2}{\delta^2}\right]^{-(\nu+1)/2}, \quad \beta = 0, \tag{4}$$

where $K_a(b)$ is the modified bessel function of the third kind (Abramowitz and Stegun, 1964) evaluated in b with (fractional) order a, and $\boldsymbol{\theta}^* = (\mu, \delta, \beta, \nu)'$ contains the location $\mu \in \Re$, scale $\delta > 0$ and the two shape parameters $\beta \in \Re$ and $\nu > 0$.

For $\beta = 0$, GHSKT reduces to the Student's t distribution with density of equation (4) and to the Gaussian distribution if also $\nu = \infty$. The dynamic model for these two special cases is detailed in Creal et al. (2013) and Harvey (2013) to which we refer for further details. For the rest of the paper, we analyze the case $\beta \neq 0$.

The first two central moments of the GHSKT distribution detailed in Aas and Haff (2006) are given by:

$$\mathbb{E}(Y) = \mu + \frac{\beta\delta^2}{\nu-2}, \tag{5}$$

$$\mathrm{Var}(Y) = \frac{2\beta^2\delta^4}{(\nu-2)^2(\nu-4)} + \frac{\delta^2}{\nu-2}, \tag{6}$$

where we note that the scale enters the mean of Y. This implies that constructing a filter for δ (i.e., $\delta_t = \psi(\delta_{t-1}, y_{t-1})$) would prevent a clear interpretation in terms of volatility estimates. Therefore the implementation of the GHSKT distribution detailed in Aas and Haff (2006) is not immediate in time–varying volatility contexts.[3] To this end, we impose the two moment conditions $\mathbb{E}(Y) = 0$ and

[3]See also Mencía and Sentana (2004) for similar arguments applied to the multivariate Generalized Hyperbolic case.

$\text{Var}(Y) = 1$, which are fulfilled by:

$$\mu = -\frac{\beta\delta^2}{\nu - 2}, \tag{7}$$

$$\delta = \left(\frac{2\bar{\beta}^2}{(\nu-2)^2(\nu-4)} + \frac{1}{\nu-2}\right)^{-1/2}, \tag{8}$$

where $\bar{\beta} = \beta\delta$. Now we can build the mapping $(\bar{\beta}, \nu) \to (\mu, \delta, \beta, \nu)$, or equivalently $\boldsymbol{\theta} \to \boldsymbol{\theta}^*$, such that for any value of $\bar{\beta} \in \Re$ and $\nu > 4$, the new density $f(y; \boldsymbol{\theta})$ satisfies the two moments conditions we require. Note that the mapping $\boldsymbol{\theta} \to \boldsymbol{\theta}^*$ is bijective, and the associated 4×2 Jacobian matrix, $\mathcal{J}_{\boldsymbol{\theta}}(\boldsymbol{\theta}) = \partial\boldsymbol{\theta}^*/\partial\boldsymbol{\theta}$, has elements:

$$\frac{\partial\delta}{\partial\bar{\beta}} = -\frac{2\bar{\beta}}{(\nu-2)^2(\nu-4)}\xi, \tag{9}$$

$$\frac{\partial\delta}{\partial\nu} = -\frac{1}{2}\left[\frac{-2\bar{\beta}^2[2(\nu-2)(\nu-4)+(\nu-2)^2]}{(\nu-2)^2(\nu-4)} - \frac{1}{(\nu-2)^2}\right]\xi, \tag{10}$$

where $\partial\beta/\partial\bar{\beta} = \left(\delta - \bar{\beta}\frac{\partial\delta}{\partial\bar{\beta}}\right)/\delta^2$, $\partial\beta/\partial\nu = -(\bar{\beta}/\delta^2)(\partial\delta/\partial\nu)$, $\partial\mu/\partial\bar{\beta} = -\left(\delta + \bar{\beta}\frac{\partial\delta}{\partial\bar{\beta}}\right)/(\nu-2)$, $\partial\mu/\partial\nu = -\bar{\beta}\left[\left(\frac{\partial\beta}{\partial\nu}(\nu-2) - \delta\right)/(\nu-2)^2\right]$, $\partial\nu/\partial\bar{\beta} = 0$, $\partial\nu/\partial\nu = 1$, and $\xi = \left(\frac{2\bar{\beta}^2}{(\nu-2)^2(\nu-4)} + \frac{1}{\nu-2}\right)^{-3/2}$. The Jacobian matrix, $\mathcal{J}_{\boldsymbol{\theta}}(\boldsymbol{\theta}) = \partial\boldsymbol{\theta}^*/\partial\boldsymbol{\theta}$, will be necessary for the derivation of the model extensions in Section 4. We also report the score of the GHSKT distribution with respect to the $\boldsymbol{\theta}^*$ parameterization, i.e. $\nabla_{\boldsymbol{\theta}^*}(y; \boldsymbol{\theta}^*) = \partial \log p(y; \boldsymbol{\theta}^*)/\partial\boldsymbol{\theta}^*$:

$$\frac{\partial \log p(y;\boldsymbol{\theta}^*)}{\partial\mu} = -\beta(\beta(y-\mu)\ell + 1) + \frac{(\nu+1)(y-\mu)}{2(\delta^2+(y-\mu)^2)}, \tag{11}$$

$$\frac{\partial \log p(y;\boldsymbol{\theta}^*)}{\partial\delta} = \frac{\nu}{\delta} + \beta^2\delta\ell - \frac{\delta(\nu+1)}{2(\delta^2+(y-\mu)^2)}, \tag{12}$$

$$\frac{\partial \log p(y;\boldsymbol{\theta}^*)}{\partial\beta} = \frac{\nu+1}{2\beta}\text{sgn}(\beta) + \ell\beta(\delta^2+(y-\mu)^2) + (y-\mu), \tag{13}$$

$$\frac{\partial \log p(y;\boldsymbol{\theta}^*)}{\partial\nu} = \frac{\log 2}{2} + \log\delta + \frac{\log|\beta|}{2} + \frac{1}{2}\zeta - \frac{1}{2}\Psi\left(\frac{\nu}{2}\right) - \frac{\log(\delta^2+(y-\mu)^2)}{4}, \tag{14}$$

where $\text{sgn}(\cdot)$ denotes the sign function, $\Psi(\cdot)$ denotes the digamma function, and

$$\ell = \frac{\frac{(\nu+1)}{2z}\left[\mathcal{I}_{-(\nu+1)/2}(z) + \mathcal{I}_{(\nu+1)/2}(z)\right] + \mathcal{I}_{-(\nu+1)/2}(z) - \mathcal{I}_{(\nu-1)/2}(z)}{z\left[\mathcal{I}_{-(\nu+1)/2}(z) - \mathcal{I}_{(\nu+1)/2}(z)\right]}, \tag{15}$$

$$\zeta = \frac{\dot{K}^{(\nu)}_{(\nu+1)/2}(z)}{K_{(\nu+1)/2}(z)z}, \tag{16}$$

with $\mathcal{I}_r(z)$ being the modified Bessel function of the first kind with order r and $\dot{K}^{(\nu)}_{(\nu+1)/2}(z)$ being the derivative of the modified bessel function of the third kind evaluated in $z = \sqrt{\beta^2(\delta^2 + (y-\mu)^2)}$ with respect to its order.[4] It follows that the score with respect to the $\boldsymbol{\theta}$ parametrization, $\nabla_{\boldsymbol{\theta}}(y;\boldsymbol{\theta}) = \partial \log p(y;\boldsymbol{\theta})/\partial\boldsymbol{\theta}$, is obtained as:

$$\nabla_{\boldsymbol{\theta}}(y;\boldsymbol{\theta}) = \mathcal{J}_{\boldsymbol{\theta}}(\boldsymbol{\theta})'\nabla_{\boldsymbol{\theta}^*}(y;\boldsymbol{\theta}^*). \tag{17}$$

The quantity $\nabla_{\boldsymbol{\theta}}(y;\boldsymbol{\theta})$ will be used to model the higher order moments of the crypto–currencies financial time series.

From equation (1), exploiting the $\boldsymbol{\theta}$ parametrization of the GHSKT distribution, it follows that:

$$Y_t|\mathbf{y}_1^{t-1} \sim \mathcal{GHSKT}(\varsigma, \sigma_t, \bar{\beta}, \nu), \tag{18}$$

where $\sigma_t = \exp(h_t)$ is the volatility at time t, and $\mathbf{y}_1^{t-1} = (y_1, \ldots, y_{t-1})'$ denotes the history of the process up to time $t-1$, see Theorem 1(c) of Blæsild (1981). According to equation (18), the two moment conditions $\mathbb{E}_{t-1}(Y_t) = \varsigma$ and $\mathrm{Var}_{t-1}(Y_t) = \sigma_t^2$ hold.[5] The score with respect to ς and σ_t is easily derived as follow:

$$\frac{\partial \log p(y_t;\varsigma,\sigma_t,\bar{\beta},\nu)}{\partial\varsigma} = -\frac{1}{\sigma_t}\nabla_\varepsilon(\varepsilon_t;\varsigma,\sigma_t,\bar{\beta},\nu), \tag{19}$$

$$\frac{\partial \log p(y_t;\varsigma,\sigma_t,\bar{\beta},\nu)}{\partial\sigma_t} = -\frac{1}{\sigma_t}(1+\varepsilon_t\nabla_\varepsilon(\varepsilon_t;\varsigma,\sigma_t,\bar{\beta},\nu)), \tag{20}$$

where $\nabla_\varepsilon(\varepsilon_t;\varsigma,\sigma_t,\bar{\beta},\nu) = \partial \log p(\varepsilon_t;0,1,\bar{\beta},\nu)/\partial\varepsilon_t$, is the derivative of the log–density of a standardized GHSKT distribution with respect to ε_t, given by:

$$\nabla_\varepsilon(\varepsilon_t;\varsigma,\sigma_t,\bar{\beta},\nu) = \beta\,(1+\beta(\varepsilon_t-\mu)\ell) - \frac{(\nu+1)(\varepsilon_t-\mu)}{2(\delta^2+(\varepsilon_t-\mu)^2)}, \tag{21}$$

where ℓ is given in equation (15) with $z = \varepsilon_t$, and β, μ and δ are computed from $\bar{\beta}$ according to the mapping $(\bar{\beta},\nu) \to (\mu,\delta,\beta,\nu)$.

[4] See Mencía and Sentana (2004) for an analytical derivation of this quantity

[5] We indicate by $\mathbb{E}_{t-1}[Y_t]$ the expectation $\mathbb{E}[Y_t|\mathbf{y}_1^{t-1}]$

2.2. The filter ϕ

Let $u_t^h = \partial \log p(y_t; \varsigma, \sigma_t, \bar{\beta}, \nu)/\partial h_t = (\partial h_t/\partial \sigma_t)(\partial \log p(y_t; \varsigma, \sigma_t, \bar{\beta}, \nu)/\partial \sigma_t)$ be the score of the conditional distribution $(Y_t|\mathbf{y}_1^{t-1})$, with respect to the log–volatility h_t, following Creal et al. (2013) and Harvey (2013), we build the following Score–Driven filter for the log–volatility level h_t:

$$h_t = \phi(h_{t-1}, y_{t-1}), \tag{22}$$

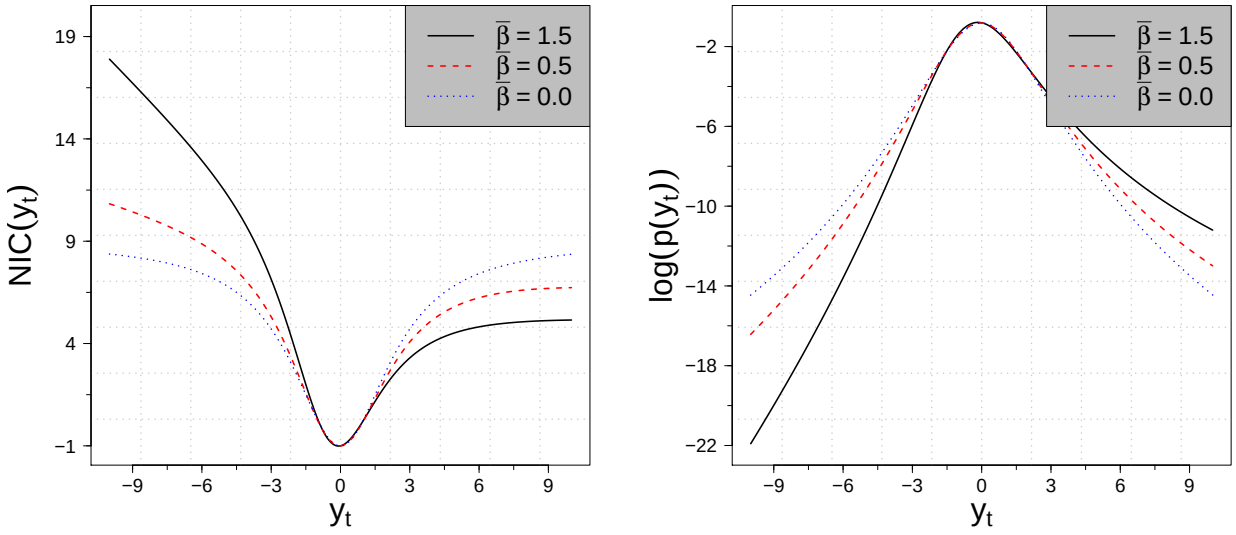
$$= \kappa^h(1 - \beta^h) + \alpha^h u_{t-1}^h + \beta^h h_{t-1}, \tag{23}$$

where $\kappa = \mathbb{E}[h_t]$ (since u_t is a Martingale Difference with respect to $(Y_t|\mathbf{y}_1^{t-1})$ for all t) is the unconditional log–volatility level, $|\beta^h| < 1$ is an autoregressive parameter controlling the persistence of the log–volatility process, and $\alpha^h > 0$ is the parameter associated to the score which determines the step of the log–volatility update. Positiveness of α^h is assumed in order to not distort the signal given by the score, see Creal et al. (2013).[6]

2.3. The volatility reaction to past returns

Engle and Ng (1993) introduced an important tool for time–varying volatility models named News Impact Curve (NIC). NIC gives insight about the reaction of future volatility levels to past observations via a graphical representation. Formally, in equation (22) we set h_t and h_{t-1} to the unconditional level κ^h, and plot the function $NIC(y_t) = \kappa^h + \alpha^h u_t^h$ for different values of y_t. Figure 1a depicts the NIC for $\kappa^h = 0$, $\alpha^h = 1$ and different values of the shape parameter $\bar{\beta}$, Figure 1b reports the associated log densities. As noted by Harvey and Sucarrat (2014) and Harvey and Lange (2017), the volatility reaction induced by Score–Driven models defined with a skewed conditional density is asymmetric and can represent the leverage effect phenomenon for selected values of the

[6] In the context of Score–Driven models, the forcing variable u_t is sometimes scaled by a positive quantity that accounts for its variance. For instance, $(\mathbb{E}[u_t^2])^{-1/2}u_t$ would imply unit variance of the resulting forcing variable of the filter. However, unfortunately the variance of the score, $\mathbb{E}[u_t^2]$, is not available in closed form for the GHSK distribution. In a related context, Lucas et al. (2017) replaced $\mathbb{E}[u_t^2]$ with the information matrix of a Student's t distribution instead of that of a GHSKT distribution. We tried with a similar approach and we find results to be qualitatively the same in our application on crypto–currency time–series.



(a) NIC **(b)** Log PDF

Figure 1: *Panel (a) reports the NIC for the volatility of the score–driven GHSKT model when* $\alpha^h = 1$ *and* $\kappa^h = 0$ *for selected values of* $\bar{\beta}$ *and* $\zeta = 0$, $\sigma = 1$, $\nu = 7$. *Panel (b) reports the associated log probability density functions.*

skewness parameter. However, even if the volatility reaction is always asymmetric for $\beta \neq 0$, the direction of the asymmetry depends on the sign of β. Indeed, for $\beta > 0$, the density is right skewed and we observe the usual leverage effect, i.e., past negative shocks increase the future volatility more than past positive shocks. Differently, when $\beta < 0$, the density is left skewed and the volatility reaction is opposed as to the previous case. This behavior is explained by noting that when $\beta > 0$ it is more unlikely to sample negative observations. Hence, when this event occurs, it is natural to have a stronger signal of increasing volatility compared to the sample of a positive observation which is indeed more likely. Obviously, the reverse holds true. From Figure 1a we also observe the winsorization property of the Score–Driven filter for finite values of ν, see Harvey (2013). However, different from the filter implied by the Student's t Score–Driven model ($\beta = 0$ in Figure 1a), in our case we observe that winsorization occurs only for positive or negative observations according to the

direction of the skewness. Specifically, we have that:

$$\lim_{y\to\infty} u_t = \begin{cases} \infty, & \text{if } \bar{\beta} < 0 \\ \text{const}, & \text{if } \bar{\beta} \geq 0. \end{cases} \tag{24}$$

$$\lim_{y\to-\infty} u_t = \begin{cases} \text{const}, & \text{if } \bar{\beta} \leq 0 \\ \infty, & \text{if } \bar{\beta} > 0 \end{cases} \tag{25}$$

which follows from the tail behaviour of the GHSKT. Indeed, as reported by Aas and Haff (2006), the heaviest tail of the GHSKT distribution decays as:

$$p(y) \sim \text{const}|y|^{-\nu/2-1} \quad \text{when} \quad \begin{cases} \beta < 0, & \text{and } y \to -\infty \\ \beta > 0, & \text{and } y \to +\infty, \end{cases} \tag{26}$$

while the lightest as

$$p(y) \sim \text{const}|y|^{-\nu/2-1}\exp(-2|\beta y|) \quad \text{when} \quad \begin{cases} \beta < 0, & \text{and } y \to +\infty \\ \beta > 0, & \text{and } y \to -\infty, \end{cases} \tag{27}$$

where we recall that $\beta = \bar{\beta}/\delta$.

3. Finite Sample Properties of the Maximum Likelihood Estimator

In this section we investigate the finite sample properties of the Maximum Likelihood Estimator (MLE) for the Score–Driven GHSKT model introduced in the previous section. Since crypto–currencies financial time–series is a quite recent kind of data, the length of the returns series is typically short. Hence, we focus on realistic sample sizes and fix $T = \{500, 700, 1000, 2000\}$. We generate $B = 10000$ samples of length T from the model detailed in the previous section. The conditional distribution parameters are fixed to $\varsigma = 0$, $\bar{\beta} = 1.5$, $\nu = 7.0$, while the coefficients driving the volatility dynamic are fixed to $\kappa^h = 0$, $\alpha^h = 0.04$, and $\beta^h = 0.94$. These parameters' value reflect those we find in the empirical study reported in Section 5 for the same series. We perform a usual Monte Carlo experiment which consists on iterating B times the following step: i) sample a

	κ^h	α^h	β^h	ς	$\bar{\beta}$	ν
			$T = 500$			
BIAS	0.003	0.001	-0.077	0.000	1.099	1.763
RMSE	0.043	0.019	0.205	0.043	4.109	5.130
SD	0.042	0.019	0.190	0.043	3.959	4.817
			$T = 750$			
BIAS	0.001	0.001	-0.042	0.000	0.477	0.925
RMSE	0.017	0.015	0.139	0.035	1.440	2.723
SD	0.017	0.015	0.133	0.035	1.359	2.562
			$T = 1000$			
BIAS	0.000	0.001	-0.026	0.000	0.318	0.635
RMSE	0.010	0.012	0.100	0.030	0.990	2.027
SD	0.010	0.012	0.096	0.030	0.938	1.926
			$T = 2000$			
BIAS	0.000	0.000	-0.009	0.000	0.147	0.307
RMSE	0.004	0.008	0.037	0.021	0.520	1.176
SD	0.004	0.008	0.036	0.021	0.499	1.135

Table 1: *This table reports the results for the finite sample properties of the Maximum Likelihood Estimator for selected values of the parameters and different sample sizes* $T = \{500, 750, 1000, 2000\}$. *Results are based on* $B = 10000$ *replicates. The coefficients' value are chosen in order to represent those found in the empirical study on crypto–currencies:* $\kappa^h = 0$, $\alpha^h = 0.04$, $\beta^h = 0.94$, $\varsigma = 0$, $\bar{\beta} = 1.5$, $\nu = 7.0$. *The table reports the bias (BIAS) and root mean squared error (RMSE) between the real and the estimated parameters, as well as the standard deviation (SD) of the estimates for different sample sizes* T.

time–series of observations from the model, ii) estimate the coefficients via MLE, and iii) store the estimated coefficients. Starting values are chosen using the same grid search procedure we employ for the empirical analysis.

Table 1 reports the bias (BIAS), root mean squared error (RMSE) between the real and the estimated parameters, as well as the standard deviation (SD) of the estimates for different sample sizes T. Clearly, for the bias going to zero we recover the equivalence between RMSE and SD. Results show that even with a small sample size, ($T = 500$), the coefficients driving the volatility dynamic, κ^h, α^h, and β^h, are well estimated via MLE. However, the conditional density parameters, ς, $\bar{\beta}$, and ν, require at least $T = 750$ observations in order to be accurately estimated. For bigger sample sizes, $T = 1000$ and $T = 2000$, all the parameters are very accurately estimated according to our experiment. Our results indicate that the bias of the estimator as well as the standard deviation are going to zero as long as the sample size increases.

4. Model Extensions

In this section we propose different extensions for the model presented in Section 2. Specifically, we aim to incorporate: i) long memory in the volatility process, ii) leverage effect, and iii) time–varying higher order moments. These extensions are then used in the empirical analysis in order to investigate the properties of crypto–currencies financial time–series. Indeed, given the novelty of these time–series for the financial econometric literature, we believe that the study of well known volatility features (long–memory and leverage effect) as well as conditional density characteristics (higher order moments) is worth doing.

4.1. Long Memory

Long memory of the volatility process is a very well–known feature of financial returns, see Andersen et al. (2001). The time–series literature developed three main methodologies to incorporate this feature in dynamic models which are: i) fractional integration (Palma, 2007), ii) multiple components models (Andersen et al., 2006), and iii) via the so–called heterogeneous autoregressive framework introduced by Corsi (2009). Fractional integrated dynamic volatility models have first been considered by Baillie et al. (1996), and recently extended by Janus et al. (2014, 2016) to the Score–Driven class of models. The first dynamic model with multiple volatility components has been proposed by

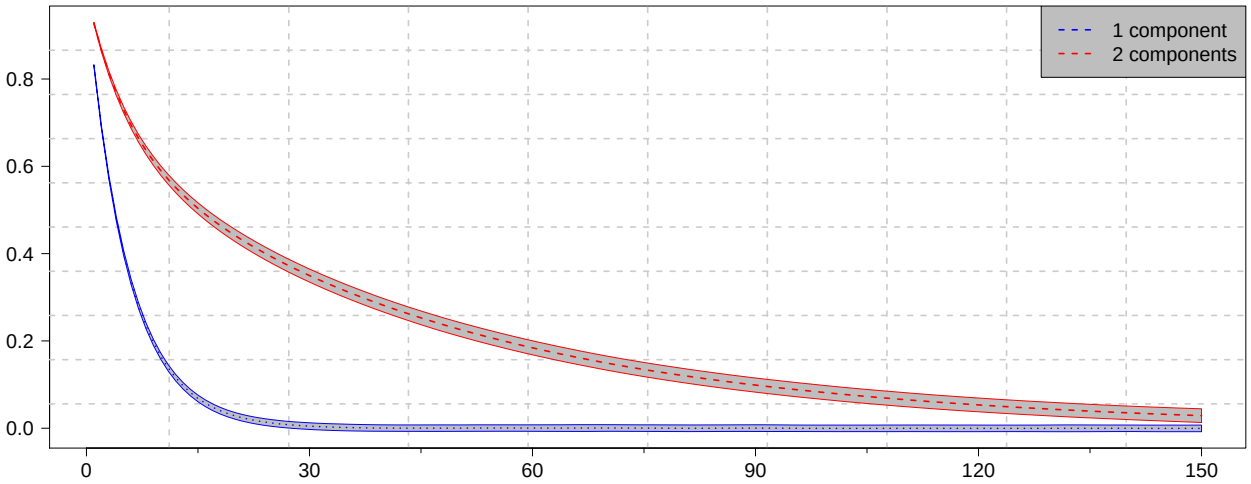


Figure 2: *Autocorrelation function evaluated for one (dotted, blue) and two (dashed, red) components ($L = 2$) GHSKT volatility model. One standard deviation confidence bands are reported in gray. Parameter values driving the volatility dynamic are: $\kappa^h = 0$, $\alpha^h = 0.06$, $\beta^h = 0.84$ for the one component model and: $\kappa^h = 0$, $\alpha_1^h = 0.04$, $\alpha_2^h = 0.06$, $\beta_1^h = 0.98$, and $\beta_2^h = 0.84$ for the two component model. Autocorrelation coefficients and the associated confidence bands are evaluated by simulation averaging over* 1000 *samples of length* $T = 100000$.

Engle and Lee (1999); recently Harvey and Sucarrat (2014) proposed a multiple component Score–Driven volatility model.[7] The Heterogeneous Autoregressive (HAR) model of Corsi (2009) has been specifically developed to model realized volatility time–series, even if its applicability clearly includes a wider range of time–series. From an unreported analysis we found that the multiple component approach, when implemented with the model of Section 2, reports better in sample results than the two alternative aforementioned approaches.[8] The inclusion of long memory via the multiple components is straightforward, essentially we rewrite equation (22) as:

$$h_t = \kappa^h + \sum_{l=1}^{L} h_{t,l}, \tag{28}$$

$$h_{t,l} = \beta_l^h h_{t-1,l} + \alpha_l^h u_{t-1}^h, \tag{29}$$

[7] See also Alizadeh et al. (2002) for an application of this methodology in the Stochastic Volatility framework.

[8] Results are available upon request to the first author.

where L is the number of components and $h_{1,l} = 0$ and $\beta_l^h > \beta_{l-1}^h$, are imposed for identifiability purposes. Namely, the log–volatility process is now defined as a sum of autoregressive processes with different autoregressive parameters and same forcing variable u_t^h. The process $\{h_t, t > 0\}$ is still weekly stationary[9] as long as $|\beta_l^h| < 1$, since $\mathbb{E}[h_{t,l}] = 0$ for all $l = 1, \ldots, L$. Clearly, for $L = 1$ we recover the model of Section 2 with a slightly different parametrisation.

Figure 2 compares the autocorrelation function of the two components model ($L = 2$) with that of the simpler one component model ($L = 1$). The decay rate of the multiple component model is evidently lower then the decay rate of the one component model which is indeed exponential.

4.2. Leverage Effect

Leverage effects refer to the asymmetric reaction of the volatility process to past positive and negative returns, a stylised fact which dates back to Crane (1959). This phenomenon has first been investigated by Black (1976) and Christie (1982) and associated to the capital structure of the firms in the spirit of the Modigliani and Miller (1958) framework. The evidence suggests that, when prices fall, volatility increases more than when prices increase. The economic justification is that a decrease in the equity value of the firm implies an increase in the leverage ratio,[10] and hence in the riskiness of the firm. This empirical evidence is also supported by the fact that foreign exchange currencies do not usually present leverage effect. However, nowadays it is more common to attribute the asymmetric reaction of the volatility to different sources, such as the different reaction of investors to past positive/negative news.

Even if nowadays the original economic interpretation of the leverage effect in terms of the changes in the capital structure of a firm has been disproved (Figlewski and Wang, 2000), the asymmetric reaction of volatility to past positive/negative returns remains a well documented stylized fact, see for example McNeil et al. (2015). In Section 2.3, we detailed how the inclusion of a skewed conditional density in Score–Driven volatility models naturally induces an asymmetric NIC. However, this kind

[9] Provided that $\mathbb{E}[(u_t^h)^2] < \infty$.

[10] Defined as the financial debt over the equity value of a firm.

of asymmetry is principally related to the parametric formulation of the model, i.e., how likely it is to sample from a particular region of the support. Differently, leverage effect is usually related to the behaviour of market participants.

A Score–Driven volatility model with the inclusion of leverage effect has first been introduced by Harvey (2013). Following Harvey's implementation, we rewrite equation (22) as:

$$h_t = \kappa^h(1-\beta^h) + \alpha^h u_{t-1}^h + \beta^h h_{t-1} + \gamma^h \text{sgn}(-(y_{t-1}-\varsigma))(u_{t-1}+1), \tag{30}$$

where γ^h is an additional parameter and $\text{sgn}(\cdot)$ is the sign function.[11] When $\gamma^h > 0$, we observe a higher increase of the volatility after negative (excess) returns. If $\gamma^h < \alpha^h$, we observe a decrease of the volatility after positive (excess) returns, i.e., the filter (30) can represent those situations when the volatility decreases after capital gains from investors. Unfortunately, due to the inclusion of the leverage effect, the volatility process defined in equation (30) is no longer updated by a Martingale Difference (MD). Indeed, if we write $g(y_t) = \alpha^h u_t^h + \gamma^h \text{sgn}(-(y_t-\varsigma))(u_t+1)$, we note that $\mathbb{E}_{t-1}[g(y_t)] = \gamma^h \varpi$, where $\varpi = 2P(\varepsilon_t < 0) - 1$, and $\varepsilon = (y_t - \varsigma)/\sigma_t$ is independently distributed as a standardized GHSKT with shape parameters $\bar{\beta}$ and ν. Clearly, when $\bar{\beta} = 0$, the distribution is symmetric and $\varpi = 0$, which implies the MD condition of $g(y_t)$. Since we want to keep the moment condition $\mathbb{E}[h_t] = \kappa^h$, the filter in equation (30) is modified as:

$$h_t = \kappa^h(1-\beta^h-\gamma^h\varpi) + \alpha^h u_{t-1}^h + \beta^h h_{t-1} + \gamma^h \text{sgn}(-(y_{t-1}-\varsigma))(u_{t-1}+1). \tag{31}$$

Figure 3 shows the implication of the leverage effect inclusion for the NIC for different values of $\gamma = \{0, 0.3, 0.6\}$. We set $\alpha^h = 1$, $\kappa^h = 0$, $\bar{\beta} = -0.5$ and $\nu = 7.0$. Since $\bar{\beta} < 0$, the conditional density is negatively skewed. Consequently, as detailed in Section 2.3, volatility reacts more to positive than negative past returns, this behaviour is represented in Figure 3 for $\gamma = 0$. However, for values of $\gamma > 0$, the direction of the asymmetry is allowed to change. Indeed, in Figure 3, when $\gamma = 0.3$ or $\gamma = 0.6$, we recover the usual leverage effect, i.e., the volatility increases more after negative past

[11] The use of $(u_{t-1}+1)$ instead of just u_{t-1} is due to the fact that omitting the $+1$ term would imply a jump of the NIC at ς.

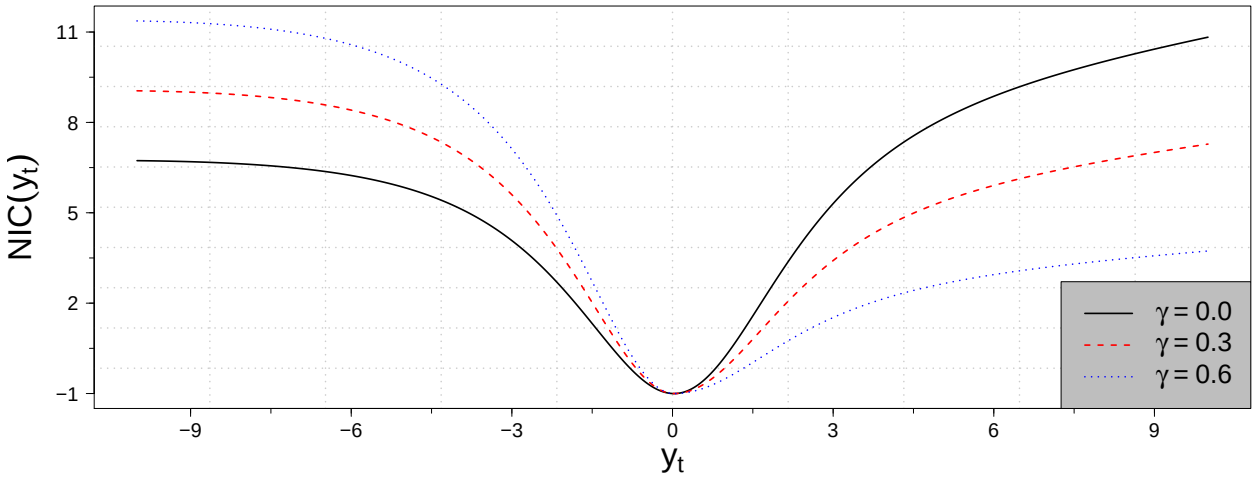


Figure 3: *NIC with the inclusion of the leverage effect component for different values of γ: solid (black) line $\gamma = 0$; dashed (red) line $\gamma = 0.3$; dotted (blue) line $\gamma = 0.6$. The other parameters as set as follows: $\alpha^h = 1$, $\kappa^h = 0$, $\bar{\beta} = -0.5$ and $\nu = 7.0$.*

returns. This means that, independently from the sign of $\bar{\beta}$, the filter reported in equation (31) is able to incorporate any kind of asymmetries in the volatility update via the inclusion of the additional parameter γ^h.

4.3. Time–varying Higher Order Moments

The inclusion of time–varying higher order moments for financial time–series dates back to the Autoregressive Conditional Density model of Hansen (1994). Extensions include the Independent Factor Autoregressive Conditional Density model of Ghalanos et al. (2015), the Autoregressive Conditional Kurtosis model of Brooks et al. (2005) among others. With regards to Score–Driven models, the inclusion of time–varying higher order moments for financial time–series has been investigated by Lucas and Zhang (2016) and Bernardi and Catania (2016) for the univariate and multivariate frameworks, respectively.

Before the development of Score–Driven models, the modeling of higher order moments was performed via ad hoc procedures. The usual methodology was that of including arbitrary transformations of past returns (say, y_{t-1}, $|y_{t-1}|$, y_{t-1}^d, for $d = 2, 3, 4$) and see if these have any sort of predictive power

for the shape parameters of the conditional distribution assumed for the returns. This strategy facilitates the tractability of the model, even if it is clearly open to a wide range of criticism. Indeed, a coherent statical answer to the choice of such transformations was missing until the development of the Score–Driven framework.

The Score–Driven methodology naturally suggests to perform filtering using the score of the conditional distribution assumed for the data. Following this general approach, any parameter in a dynamic model, for which a parametric distributional assumption is available, can be time–varying. Specifically, the quantities $u_t^{\bar{\beta}} = \partial \log p(y_t; \boldsymbol{\theta})/\partial \bar{\beta}$ and $u_t^{\nu} = \partial \log p(y_t; \boldsymbol{\theta})/\partial \nu$ collected in the vector $\nabla_{\boldsymbol{\theta}}(y_t; \boldsymbol{\theta}_t)$ reported in equation (17), can be used to build a filter for $\bar{\beta}_t$ and ν_t. Indeed, as for the volatility parameter, we can write:

$$\bar{\beta}_t = \kappa^{\bar{\beta}}(1 - \beta^{\bar{\beta}}) + \alpha^{\bar{\beta}} u_{t-1}^{\bar{\beta}} + \beta^{\bar{\beta}} \bar{\beta}_{t-1}, \tag{32}$$

for the time–varying skewness parameter, $\bar{\beta}_t$, and

$$\nu_t = \exp(\widetilde{\nu}_t) + 4, \tag{33}$$

$$\widetilde{\nu}_t = \kappa^{\nu}(1 - \beta^{\nu}) + \alpha^{\nu} u_{t-1}^{\nu} \nu_{t-1} + \beta^{\nu} \widetilde{\nu}_{t-1}, \tag{34}$$

for the time–varying shape parameter, $\widetilde{\nu}_t$. Note that, in equation (33), we have employed an exponential link function with a lower bound of 4 in order to ensure the positiveness of ν_t, and the existence of the variance of the conditional distribution $Y_t|\mathbf{y}_{1:t-1}$. It is also worth noting that, in the filter (34), we have multiplied the conditional score u_{t-1}^{ν} by ν_t, which is the derivative of the mapping $\widetilde{\nu}_t \to \nu_t$, such that the forcing variable $u_t^{\nu}\nu_t$ is equivalent to $\partial \log p(y_t; \boldsymbol{\theta})/\partial \widetilde{\nu}_t$.

5. Empirical Application

The data used in this study are the crypto–currencies closing prices. It is important to underline that the crypto–market is open 24 hours a day, seven days a week; hence for the closing price we use prices at midnight (UTC). Those data are freely available from CoinMarketcup.[12]

[12] https://coinmarketcap.com/

Since the introduction of Bitcoin in 2009, hundreds of other crypto–currencies have been created and, as of July 2017, 810 crypto–currencies exist. Some of them, though, do not represent serious attempts at establishing a foothold in the market, others are quite young with just a few useful observations. For this reason, we focus on coins with at least 700 observations, which correspond to almost 2 years of data; this leaves us with 289 series to analyze. We group each series into three categories based on the market capitalization recorded during the month of July 2017. Those categories are: i) Large Cap (top 20%), ii) Mid Cap (between 40% and 60%), and iii) Small Cap (lower 20%). Hence, we report results at the aggregate level (All), for the three categories (Large, Mid and Small Cap), and for the four major crypto–currencies: i) Bitcoin, ii) Ethereum, iii) Ripple, and iv) Litecoin. We now briefly detail these four crypto–currencies.

Bitcoin is the most popular and prominent crypto–currency based on the decentralization and cryptography. The decentralization means that the Bitcoin network is controlled and owned by all of its users, who must adhere to the same set of rules. The cryptography controls money creation (fixed to a maximum of 21 million coins) and transactions, no central bank is needed, see Nakamoto (2009). This decentralized nature offers many advantages, such as being free from government control and regulation, but critics often argue that apart from its users, there is nobody overlooking the whole system and that the value of Bitcoin is unfounded.

Despite that, since its creation from a starting price of few cents in 2010, it touched 12015 USD in December 2017 and at the time of writing is around 11850 USD.

Ethereum is a decentralized platform that runs smart contracts which facilitates online contractual agreements applications that run any possibility of downtime, censorship, fraud, or third party interference. The Ethereum also provides a crypto–currency token called Ether which can be transferred between accounts and used to compensate participant nodes for computations performed, see Ethereum (2014).

Ripple, developed by the banking industry in 2012, is a blockchain network which incorporates a payment system and a currency system known as XRP. It enables banks to send real–time international payments across them and for this reason is currently used by many banks such as

UBS, Santander, and Standard Chartered among many others, see Ripple (2012).

Litecoin has been created in 2011 and is based on the same protocol used by Bitcoin. For this reason it is often considered Bitcoin's leading rival. It has one main feature which distinguishes it from Bitcoin: it is significantly faster regarding transactions, and it is particularly attractive in time–critical situations, see Litecoin (2014).

In Subsection 5.1 we describe the main four cryptocurrencies and we report the statistical properties of their unconditional distribution. In Subsection 5.2 we compare the Score–Driven GHSKT model with the widely used GARCH model of Bollerslev (1986). Subsections from 5.3 to 5.5 extend the analysis to long memory, leverage effect, and time–varying higher order moments.

5.1. Data and unconditional distribution of crypto–currencies financial time–series

Table 2 reports some descriptive statistics for the four principal crypto–currencies: Bitcoin, Ethereum, Ripple, and Litecoin. We note that Ripple and Litecoin have the lowest minimum and the largest maximum. The means and medians of the log–returns of all the currencies are different from zero with just Bitcoin that has mean and median almost equal to zero. Only the log–returns of Bitcoin is negatively skewed, all others are positively skewed with Litecoin having the highest value. Bitcoin returns have the lowest standard deviation of the four crypto–currencies, those with the highest variation are Ripple and Ethereum: two currencies that experience an enormous growth in 2017.

In Figure 4 we report the median, standard deviation, Skewness, and Kurtosis for all the crypto–currencies and for Large, Middle, and Small cap. As the figure shows, the median (Panel a), Skewness (Panel c), and Kurtosis (Panel d) are very similar between Large, Mid, and Small cap. The Kurtosis, concentrated around 15, is quite high with extreme values near 100. The majority of the series show a positive Skewness coefficient, concentrated around 1.7, with the presence of extreme values in the positive and negative region of the order -6 to $+8$. Finally, the standard deviations (Panel b) are quite heterogeneous across Large, Mid, and Small Cap, with biggest values for the Large Cap and relatively moderate values for the Small Cap. Interestingly, crypto–currencies financial time–series also exhibit serial autocorrelation as documented by the empirical autocorrelation coefficients (not reported). We believe this phenomenon requires an extended analysis and decided to leave it for

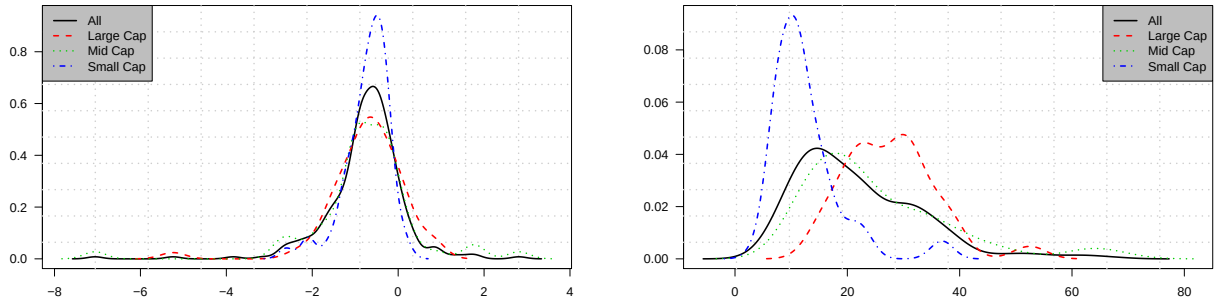
Coin	Bitcoin	Ethereum	Ripple	Litecoin
Nr. Obs.	1497	702	1412	1437
Created	03-Jan-09	01-Aug-14	01-Jul-13	01-Nov-13
Supply	21 Millions Total	18 Millions Yearly	100 Billions Total	84 Millions Total
Market Cap	200 Billions	44 Billions	9.5 Billions	5.5 Billions
Maximum	31.733	30.989	99.182	82.325
Minimun	-24.065	-28.747	-64.175	-51.942
Mean	0.002	0.134	0.037	0.019
Median	0.026	-0.870	-0.322	-0.323
Std Dev.	4.045	7.252	7.607	6.853
Skewness	-0.371	0.584	1.840	2.052
Kurtosis	11.620	5.964	34.850	33.066

Table 2: *Four larger crypto–currencies by market capitalization. The table reports the name of the coins, the number of observations, the creation date, the maximum number of coins in Million (Mil.) and Billion (Bil.) and the market capitalization as in July 2017 reported in https://coinmarketcap.com/. The Ethereum has a total supply of 18 Millions coins per year, the other three have a total prefix amount. The table also reports for each coin some summary statistics.*

future research. Consequently, we pre–filter the series using a simple autoregressive filter of order p, with p chosen according to the Akaike Information Criterion.

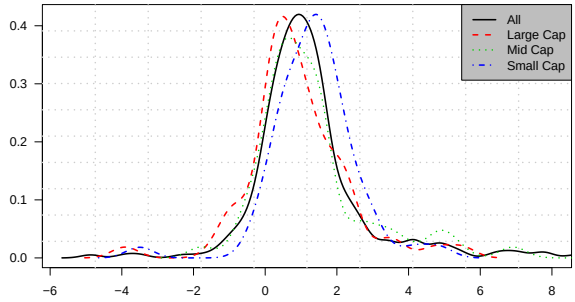
5.2. Score–Driven GHSKT vs GARCH–GHSKT

We begin by studying the evolution of the volatility process of crypto–currencies time–series and by emphasizing the importance of a robust filter for the volatility update. To this end, we compare the GHSKT Score–Driven model with a GARCH model with the same conditional distributional

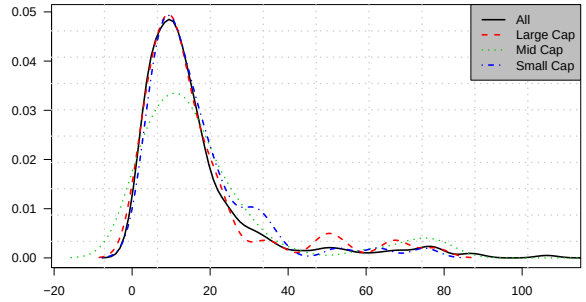


(a) Median

(b) Standard Deviation



(c) Skewness



(d) Kurtosis

Figure 4: *Kernel densities of the median (a), standard deviation (b), skewness (c) and kurtosis (d) coefficients of the full sample of crypto–currencies time–series. Kernel densities associated to the subgroups: i) Large Cap (red dashed line), ii) Mid Cap (green dotted line), and iii) Small Cap (blue dot-dashed line) are also reported. Groups are made according to the average market capitalization in USD dollar during the month of July, 2017.*

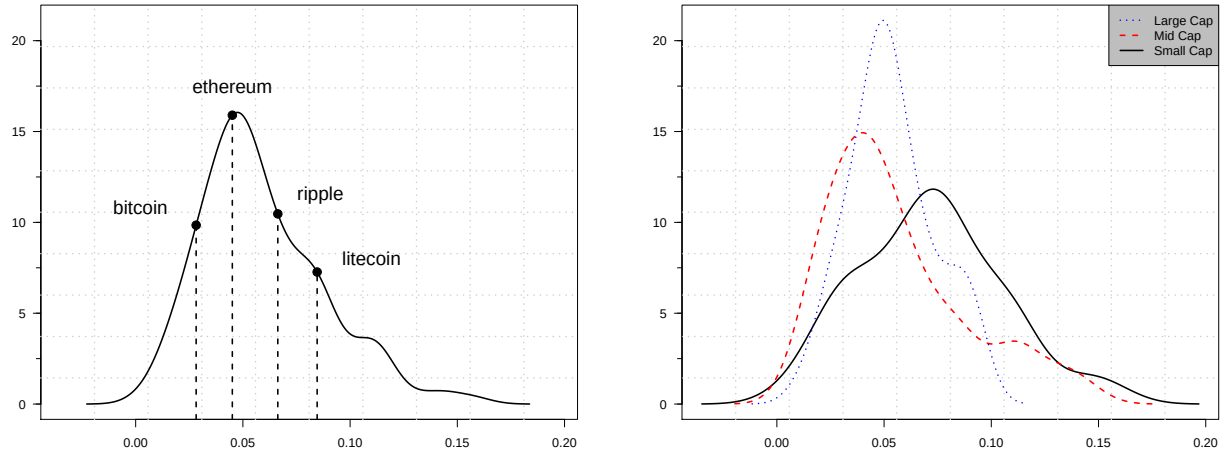
assumption, defined as

$$y_t = \varsigma + \sigma_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{GHSKT}(0, 1, \bar{\beta}, \nu), \tag{35}$$

$$\sigma_t^2 = \zeta_0 + \zeta_1 (y_{t-1} - \varsigma)^2 + \zeta_2 \sigma_{t-1}^2, \tag{36}$$

where $\zeta_i > 0$, for $i = 0, 1, 2$ and $\zeta_1 + \zeta_2 < 1$, see Bollerslev (1986). It is worth noting that the NIC implied by the filter in equation (35) is quadratic in the residual $y_t - \varsigma$, and hence different from the one implied by the Score–Driven filter; it is not robust to extreme observations. Also note that the number of parameters for the GARCH and Score–Driven model is the same such that in sample model comparison can be performed by a simple likelihood comparison. Figure 5 reports a Gaussian

kernel fitted on the likelihood percentage increases of the Score–Driven model over GARCH across the 289 crypto–currencies.[13] Panel (a) reports the percentage increase for the whole cross–section

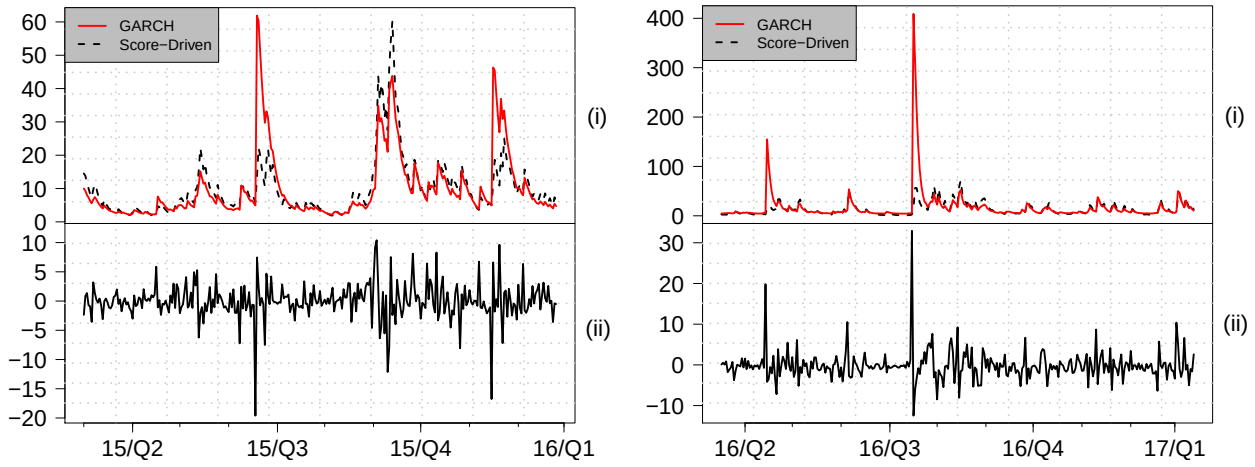


(a) All **(b)** Subgroups

Figure 5: *Kernel density estimated on the likelihood percentage increase between the GHSKT Score–Driven model and the GARCH–GHSKT model. Positive values indicate outperformance of the GHSKT Score–Driven with respect to the GARCH–GHSKT alternative. Panel (a) reports the density for all crypto–currencies (All), Panel (b) for the three groups of crypto–currencies: Large Cap, Mid Cap, and Small Cap. Groups are made according to the average market capitalization in USD dollar during the month of July, 2017.*

while Panel (b) reports for the three categories Large, Mid and Small Cap. We find that on average the GHSKT Score–Driven model reports a likelihood increase in the order of 5%. Furthermore, the GARCH always reports a lower likelihood than the Score–Driven specification. Across the four major crypto–currencies, Litecoin is the one which benefits most from the Score–Driven filter with a gain of 8.5%, then Ripple (6.6%), Ethereum (4.5%), and Bitcoin (2.8%). Looking at Figure 5a, we note that the likelihood increments vary with the size of the market cap. Specifically, likelihood increments are on average 7.2% for Small Cap, 5.6% for Mid Cap, and 5.4% for Large Cap crypto–currencies.

[13] We use a Gaussian kernel density estimator with bandwidth selected by "Silverman's rules–of–thumb" criteron (see Silverman, 1986, p. 48). Alternative kernel and bandwidth choices led to similar results.



(a) Bitcoin **(b)** Ripple

Figure 6: *Filtered conditional variances for Bitcoin (Panel (a)) and Ripple (Panel (b)) delivered by the Score–Driven GHSKT model (black dashed line) and the GARCH (red solid line). For each figure, the sub–panel (i) reports the conditional variance and sub–panels (ii) reports the log–return in percentage points.*

Interestingly, we find that for Small Cap crypto–currencies the use of a robust volatility filter is much more important than for Mid and Large Cap, hence for those series the robustness of the filter is an essential requirement. In order to show the difference implied by the robust Score–Driven volatility filter, in Figure 6 we report the filtered variances for a sub–sample of data delivered by the two models for Bitcoin (Panel (a)) and Ripple (Panel (b)). The lower panel of each figure shows the log–returns series. The plot is very clear; whenever there is an extreme observation, the GARCH filter reports a huge increase in the volatility level, even if the signal of increased volatility is fairly weak. For instance, in Figure 6b, we see that on the 15th September 2016 Ripple exhibited a log–return of roughly 30% due to a received founding of 55 USD millions from private banks.[14] The reaction of GARCH was that of increasing the conditional variance from 5.8 to 408.9, while the increase delivered by the Score–Driven specification was from 2.8 to 15.6. Indeed, after that extreme observation the

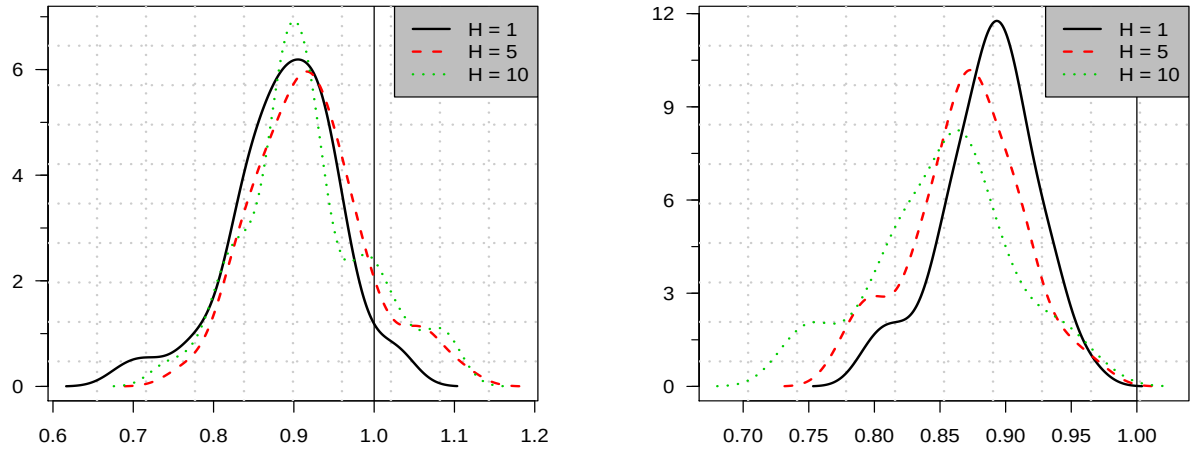
[14]Source:http://www.cnbc.com/2016/09/15/google-backed-blockchain-start-up-ripple-raises-55-million-from-big-banks.html

overall volatility level hasn't changed too much, indicating that the huge increase in volatility reported by GARCH was not justified by the data.

5.2.1. A brief overview on quantitative risk management applications

In order to note the benefits of a robust filter for the volatility process of crypto–currencies time–series, we now report a brief forecasting analysis comparing predictions delivered by the robust Score–Driven GHSKT filter and the GARCH filter. We would like to stress that an extensive out–of–sample analysis is beyond the scope of this paper which instead concentrates on the in–sample properties of the volatility of crypto–currencies time–series. The quantity of interest for this analysis is the H–step ahead Value–at–Risk (VaR) level of the crypto–currency time–series at confidence level $\alpha \in (0, 1)$. From a risk management perspective, VaR indicates the maximum possible loss excluding events that occur with probability less than α. We refer to Jorion (2006) for an extensive coverage of the topic. In statistical terms, VaR coincides with the quantile of the predictive distribution of the random variable of interest evaluated at the chosen confidence level α. In our case, the random variable of interest is defined as $Y_{T+H}|\mathbf{y}_1^T$, where T is the length of the in–sample period where model's parameters are estimated. We chose the two standard confidence levels $\alpha = 1\%$ and $\alpha = 5\%$ and three forecast horizons: i) one day ($H = 1$), ii) one week ($H = 5$), and iii) two weeks ($H = 10$).

Starting from our data set of 289 time–series, we select those with a sample size of at least 1.200 observations leaving us with 49 series. The in–sample size of the first forecast is set to $T = 600$, such that we have at least 600 predictions in order to compare the two models. Models' parameters are estimated using the first 600 observations and subsequently getting updated each time a new observation becomes available using an extending window. For both models, multi step ahead predictions ($H > 1$) of the VaR level are computed by Monte Carlo simulation using 10.000 draws from the predictive distribution. In order to compare the Score–Driven and GARCH GHSKT models we compute the quantile loss associated to each VaR prediction delivered by the two models. Given a prediction $VaR^{\alpha}_{T+H|T}$, and the realization of the crypto–currency log–return, y_{T+H}, the quantile



(a) $\alpha = 1\%$ **(b)** $\alpha = 5\%$

Figure 7: *Kernel density estimated on the ratio between the average quantile loss of the Score–Driven and GARCH GHSKT models. Only crypto–currency time–series with at least 1'200 data points have been included for this analysis. Models' parameters are estimated using the first 600 observations and subsequently getting updated each time a new observation becomes available using an extending window. Figure (a) reports results for the $\alpha = 1\%$ VaR confidence level, Figure (b) reports results for $\alpha = 5\%$. Values lower than one indicate outperformance of the predictions based on the Score–Driven volatility filter versus the GARCH alternative. The vertical bar at one indicates equal performance between the two specifications.*

loss is defined as:

$$QL^{\alpha}_{T+H|T} = (\alpha - \mathbf{1}_{\{y_{T+H} < VaR^{\alpha}_{T+H|T}\}})(y_{T+H} - VaR^{\alpha}_{T+H|T}), \tag{37}$$

where $\mathbf{1}_{\{y_{T+H} < VaR^{\alpha}_{T+H|T}\}} = 1$ if $y_{T+H} < VaR^{\alpha}_{T+H|T}$ and zero otherwise, we refer to González-Rivera et al. (2004). Quantile loss values are subsequently averaged over the forecasting horizons, the model with the lower averaged QL value is preferred.

Figure 7 reports a kernel density estimated on the ratio between the average quantile loss of the Score–Driven and GARCH GHSKT models, for both VaR levels and forecast horizons. Values lower than one indicate outperformance of the predictions based on the Score–Driven volatility filter versus the GARCH alternative. We clearly see that, for crypto–currency time–series, a robust volatility filter is essential for VaR predictions. Indeed, when $\alpha = 1\%$, for the majority of crypto–currencies

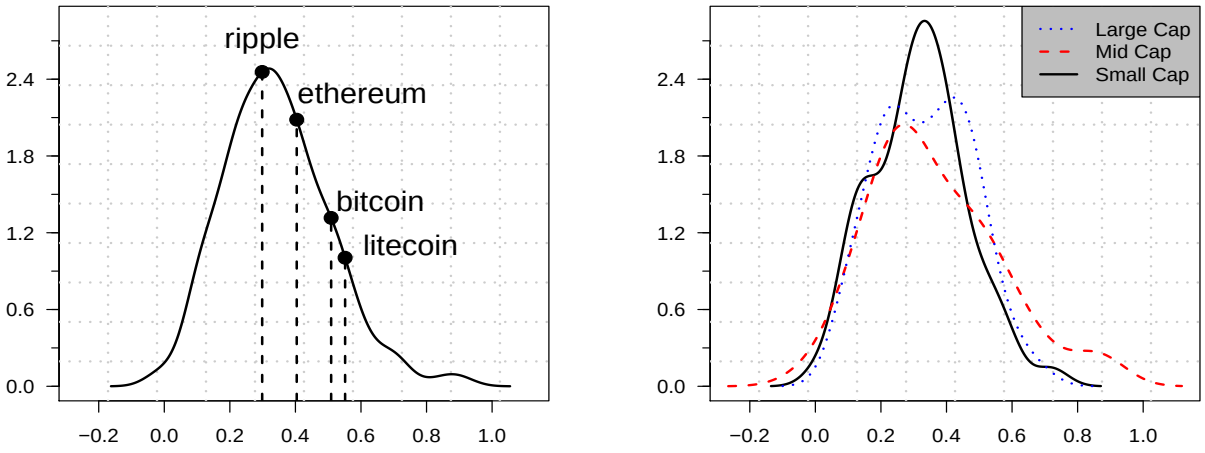
we observe the Score–Driven GHSKT model outperforms GARCH. Specifically, on average gains are: 11.1% for $H = 1$, 8.1% for $H = 5$, and 8.5% for $H = 10$. When $\alpha = 5\%$, we note that for all crypto–currencies the Score–Driven GHSKT model outperforms GARCH. In this case, gains are on average: 11.3% for $H = 1$, 13.1% for $H = 5$, and 14.8% for $H = 10$. We plan to extend these very encouraging preliminary results including the long–memory, leverage effect, and time–varying higher order moments features in a future research paper.

5.3. Long memory

From the financial econometrics literature we know that stock volatility often exhibits long memory properties, see among others Lobato and Velasco (2000) and Rossi and Santucci de Magistris (2013). Here we investigate if this feature also holds for crypto–currencies time–series. Figure 8 reports the fractional integration coefficient d estimated on the squared returns using the estimator proposed in Geweke and Porter-Hudak (1983).

Looking at Figure 8 Panel (a), the estimated d has big variation with the highest concentration around the stationarity region, $0 \leq d \leq 0.5$. Bitcoin and Litecoin are in the nonstationary region with d equal to 0.51 and 0.55 respectively, while Ripple and Ethereum have smaller d parameters equal to 0.30 and 0.40 respectively. Panel (b) of Figure 8 reports the estimated d parameters by market capitalization. The group that presents the highest values is the Mid cap. The Large cap presents bimodality around 0.2 and 0.5 suggesting that market capitalization might not be the only source of heterogeneity across crypto–currencies financial time–series. As for Panel (a), the highest concentration is around the stationarity region, $0 \leq d \leq 0.5$. We have to underline that testing for fractional integration (true long memory) versus spurious long memory as in Qu (2011) or comparing the performance of different d estimators as in Leccadito et al. (2015) is outside the scope of the paper and we leave it for further research.

As detailed in Section 4.1, our long–memory Score–Driven volatility specification does not rely on the use of a fractional integrated process. Instead, we make use of the multiple volatility component approaches as in Harvey and Sucarrat (2014), see Section 4.1. We estimate the multiple component volatility models with a different number of components ($L = 1, 2, 3, 4$) on the crypto–currencies



(a) All series

(b) Series by different market capitalization

Figure 8: *Kernel density estimated on the fractional parameter, d estimated using the time–series of squared returns. The coefficient of fractional integration is estimated using the estimator described in Geweke and Porter-Hudak (1983). Panel (a) reports the density for all crypto–currencies (All), Panel (b) for the three groups of crypto–currencies: Large Cap (blue dotted line), Mid Cap (red dashed line), and Small Cap (black solid line). Groups are made according to the average market capitalization in USD during the month of July, 2017.*

series and perform a comparison via penalized likelihood criteria. Columns 1 to 4 of Table 3 report the selected specification according to the Akaike information criterion (AIC), the Bayesian information criterion (BIC), and the Hannan–Quinn information criterion (HQ) for the four major crypto–currencies. Columns 5 to 8 of Table 3 report the percentage of times each specification is preferred for the three categories Small, Mid, and Large Cap as well as for the whole sample, All. We find that, according to the three information criteria, for Bitcoin and Ethereum one component ($L = 1$) is enough. Differently, for Ripple AIC selects three components, while BIC and HQ select $L = 2$. Similarly for Litecoin, AIC and HQ select 2 components, while BIC selects $L = 1$. Concerning the aggregated results, we find that BIC generally favours $L = 1$, AIC generally favours $L = 2$, and HQ select $L = 1$ and $L = 2$ with similar frequencies. Interestingly, we find that for Large Cap crypto–currencies a multiple component volatility model is generally more adequate than for Small and Mid Cap. Indeed, according to BIC, 35% of the Large Cap series require $L = 2$, while Mid and

#	Bitcoin	Ethereum	Ripple	Litecoin	Small Cap	Mid Cap	Large Cap	All
Panel (a): AIC								
1	x	x	–	–	35	33	25	28
2	–	–	–	x	58	61	61	65
3	–	–	x	–	7	4	14	7
4	–	–	–	–	0	2	0	0
Panel (b): BIC								
1	x	x	–	x	75	74	65	66
2	–	–	x	–	25	26	35	34
3	–	–	–	–	0	0	0	0
4	–	–	–	–	0	0	0	0
Panel (c): HQ								
1	x	x	–	–	51	56	37	44
2	–	–	x	x	49	44	61	56
3	–	–	–	–	0	0	2	1
4	–	–	–	–	0	0	0	0

Table 3: *Number of volatility components for the GHSKT Score–Driven multiple components volatility model introduced in Section 4. For the four major crypto–currencies: Bitcoin, Ethereum, Ripple, and Litecoin, the label "x" indicates the number of components chosen according to the corresponding information criteria. Panel (a) reports results for AIC, Panel (b) for BIC, and Panel (c) for HQ. Columns 5 to 8 report the percentage of times each model is chosen according to AIC for the three groups: Small Cap, Mid Cap, and Large Cap, as well as over the full sample (All).*

Small Cap series require $L = 2$ only 25% and 26% of times, respectively. Interestingly, AIC also selects models with 3 and 4 components, while these specifications are generally not selected by BIC and HQ. However, we note that AIC is more prone to select overparameterized models.

	Bitcoin	Ethereum	Ripple	Litecoin	Small Cap	Mid Cap	Large Cap	All
AIC	–	x	x	x	54	46	72	57
BIC	–	–	x	–	26	19	44	30
HQ	–	–	x	–	42	32	58	42

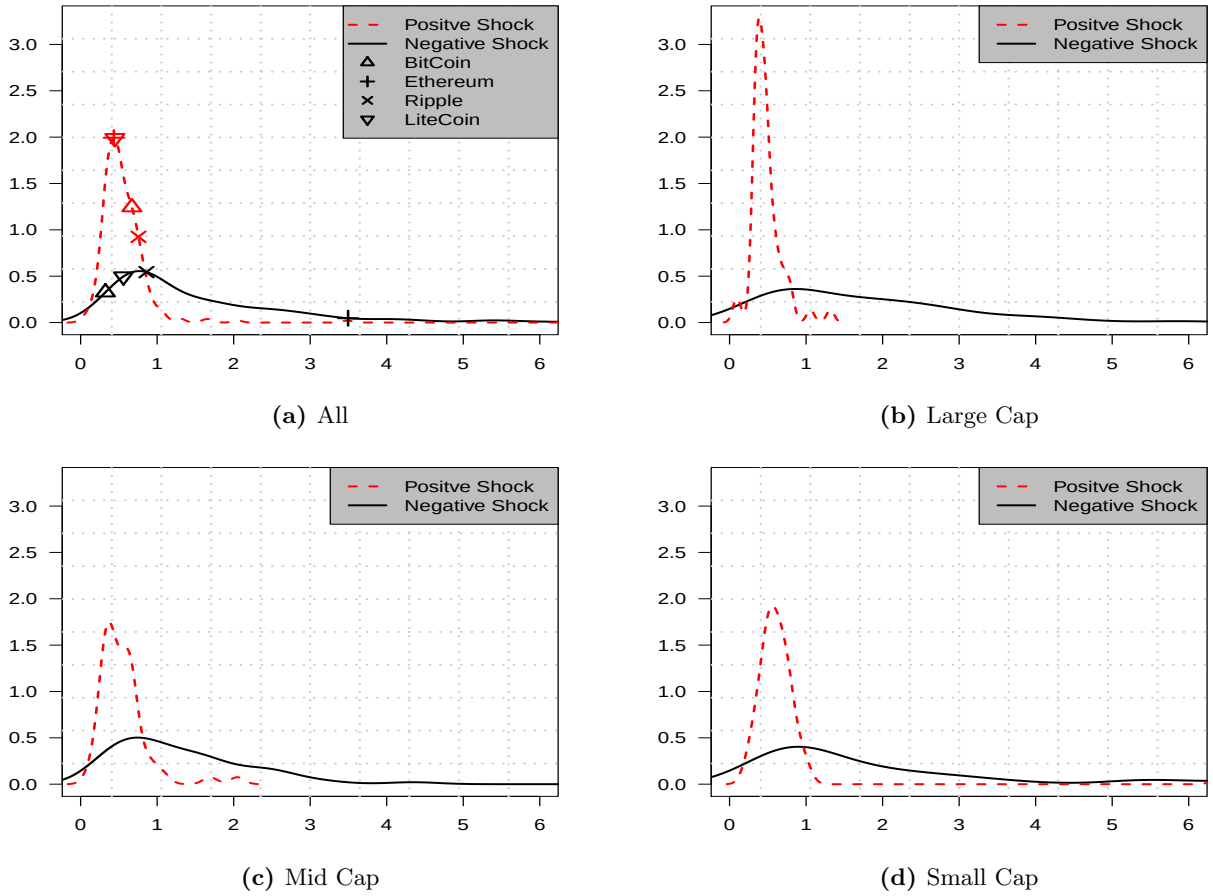
Table 4: *Comparison between models with and without the leverage component. Comparison is made according to the three information criteria: AIC, BIC, and HQ. For the four major cryptocurrencies: Bitcoin, Ethereum, Ripple, and Litecoin, the label "x" indicates preference with respect to the model with the leverage component, the label "–" indicates the opposite. Columns 5 to 8 report the percentage of times the model with the inclusion of the leverage component is preferred for the three groups: Small Cap, Mid Cap, and Large Cap, as well as over the full sample (All).*

5.4. Leverage effect

So far we find that: i) a robust filter for the volatility of crypto–currencies financial time–series is strongly required by the data, and ii) there is evidence of long memory in the volatility update for some series while for others a simpler specification is enough. In this section we aim at investigating the asymmetric reaction of the volatility process to past positive and negative shock, as well as if crypto–currencies financial time–series exhibit the so–called leverage effect, *i.e.*, if the volatility increases more after negative shocks than after positive shocks.

As reported in Section 2.3, the volatility filter implied by the Score–Driven GHSKT model already incorporates an asymmetric volatility reaction which depends on the skewness of the conditional distribution of the data. However, the direction of the asymmetries might be opposite to the empirical regularities observed in financial time–series. To this end, in Section 2.3 we propose a new filter able to incorporate different kinds of volatility reaction. As a first result, we show if there is statistical evidence in favour of the extended model. In Table 4 we report an in sample comparison between the Score–Driven GHSKT model introduced in Section 2 and the modification with the inclusion of the leverage effect introduced in Section 2.3.

Since the model extension has one additional parameter, comparison is made via the three



(a) All **(b)** Large Cap

(c) Mid Cap **(d)** Small Cap

Figure 9: *These figures show the estimated reaction of the volatility level after a negative (dotted, red) and positive (solid, black) shocks. We report the NIC evaluated at* $\varepsilon_t = -15$ *(negative shock) and* $\varepsilon_t = 15$ *(positive shock) across the sample of crypto–currencies. Random variables have been standardized in order to be comparable. Panel (a) reports the density for all crypto–currencies (All), Panel (b) for the large cap (Large Cap), Panel (c) for the middle cap (Mid Cap), and Panel(d) for the small cap (Small Cap). Groups are made according to the average market capitalization in USD during the month of July, 2017.*

information criteria AIC, BIC, and HQ. We find that, according to AIC, the extended model is preferred for Ethereum, Ripple, Litecoin, but not for Bitcoin. On the aggregate level, the extended model is preferred 72% of times for Large Cap, 46% for Mid Cap, and 54% for Small Cap. Over the full cross–section, the extended model is preferred 57% of times according to AIC. Percentages decrease

if model selection is performed via BIC and HQ. Interestingly, according to the three information criteria, Ripple requires the extended model. Looking at the differences across the three market cap categories, we note that the inclusion of the leverage effect is required more by Large Cap than by Mid and Small Cap crypto–currencies. These findings are similar to those of the analysis on the number of volatility components reported in the previous Section 5.3. That is, the volatility dynamic is heterogeneous with respect to the cryptocurrency market capitalization.

In order to investigate the magnitude of the volatility increase after a positive and negative shock, in Figure 9 we report the value of the NIC evaluated at $\varepsilon_t = -9$ and $\varepsilon_t = 9$ implied by the extended model of Section 5.3.

Very interestingly, we find that different from usual foreign exchange currencies, leverage effect has a substantial contribution to the volatility dynamic. On average, volatility increases more after negative shock than after positive shock as it usually happens in the equity market. Hence, crypto–currencies time–series incorporate the so–called leverage effect. However, we note that the extent of this behaviour is quite heterogeneous across different series. Indeed, the kernel densities reported in Figure 9 for the negative shock are much more disperse than those implied after a positive shock. Looking at the four major crypto–currencies series, we note that the reaction to past positive and negative shock is quite similar. The only exception is Ethereum for which the log–volatility reaction to past negative shock (-9) is around 3.5, while after a positive shock ($+9$) is approximately 1.

5.5. Time–varying Higher Order Moments

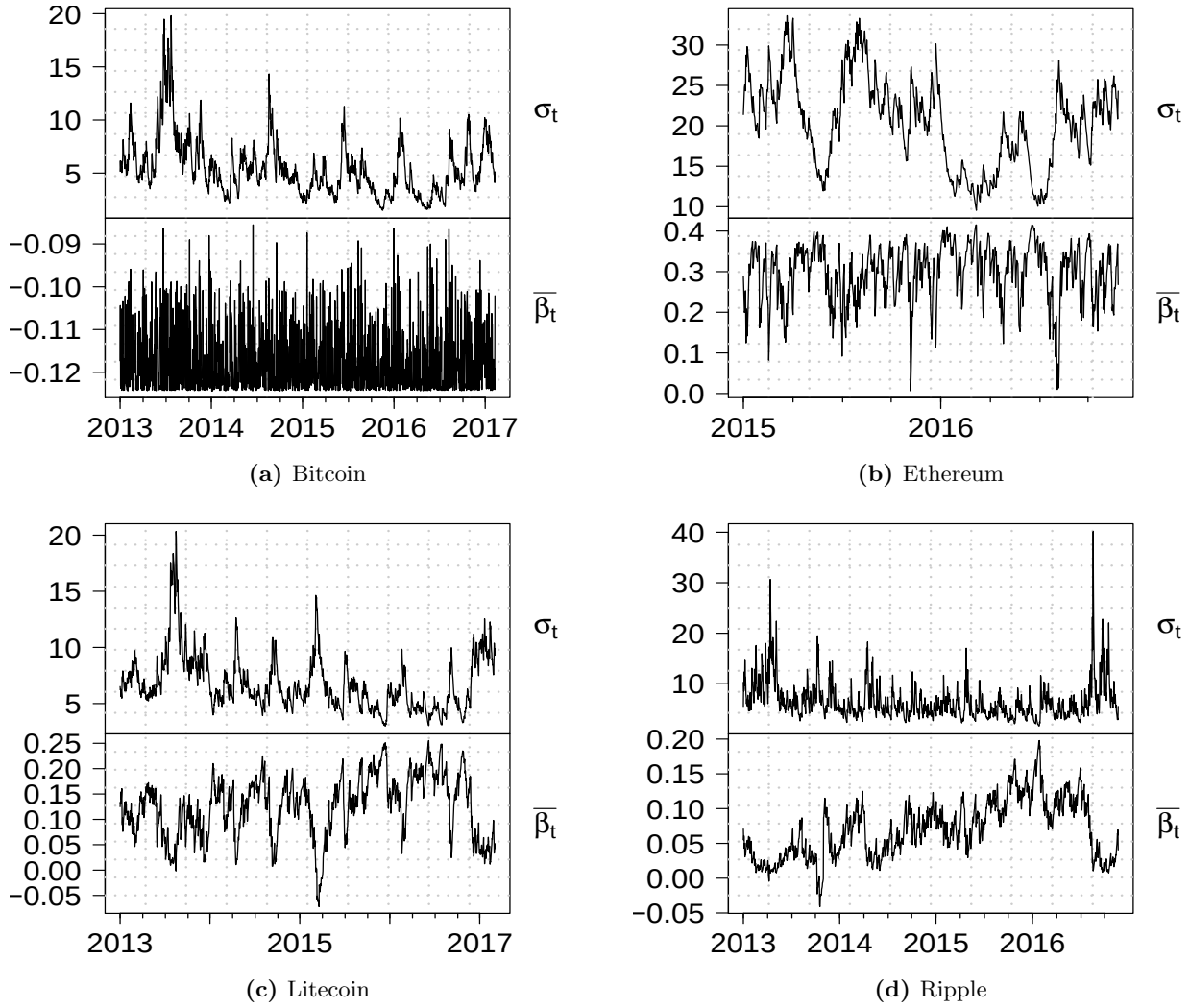
In the last part of our analysis, we investigate the presence of time variation in the higher order moments of the conditional distribution of crypto–currencies financial time–series. We follow the methodology introduced in Section 4.3 and estimate the two extended models which incorporate time–varying skewness and kurtosis. As for the previous analysis, we perform in–sample model comparison between the extended specifications and the one introduced in Section 2 using the three information criteria AIC, BIC, and HQ. Table 5 reports the percentage of times the model with the new feature is preferred over the simpler specification for the three categories: Small, Mid, and Large Cap, as well as for the full cross–section of data. Individual results for the four major crypto–currencies are also

	Bitcoin	Ethereum	Ripple	Litecoin	Small Cap	Mid Cap	Large Cap	All
Panel (a): Time–varying Skewness								
AIC	–	x	–	–	35	35	49	41
BIC	–	–	–	–	11	7	26	14
HQ	–	–	–	–	25	23	39	28
Panel (b): Time–varying Kurtosis								
AIC	–	–	–	–	0	0	0	0
BIC	–	–	–	–	0	0	0	0
HQ	–	–	–	–	0	0	0	0

Table 5: *Comparison between models with and without the inclusion of time–varying skewness and kurtosis. Comparison is made according to the three information criteria: AIC, BIC, and HQ between the plain Score–Driven GHSKT model and the extension with time–varying skewness and kurtosis. For the four major crypto–currencies: Bitcoin, Ethereum, Ripple, and Litecoin, the labels "x" indicate preference with respect to the extended model. Columns 5 to 8 report the percentage of times the extended model is preferred for the three groups: Small Cap, Mid Cap, and Large Cap, as well as over the full sample (All).*

reported.

Panel (a) of Table 5 reports results for the model with time–varying skewness. We note that there is no evidence in favour of the extended specification for the four major crypto–currencies. The only exception is Ethereum when model comparison is performed according to AIC; however, BIC and HQ favour the simpler specification without time–varying skewness. Overall, we find that the percentage of times the extended specification is preferred ranges from 14% to 41% depending on which information criterion is used. Regarding the three market capitalization categories, we still find that for Large Cap crypto–currencies, on average, the extended specification is preferred. Indeed, according to BIC, for 26% of the Large Cap crypto–currencies the model with time–varying skewness is selected, versus the 7% and 11% of times for Mid Cap and Small Cap, respectively. Even if information criteria most of the times favour the simpler specification, we find that the estimated



(a) Bitcoin

(b) Ethereum

(c) Litecoin

(d) Ripple

Figure 10: *Filtered volatility (σ_t) and skewness parameter ($\bar{\beta}_t$) for the four most capitalized crypto–currencies: Bitcoin, Ethereum, Litecoin, and Ripple over the full sample.*

score coefficient $\alpha^{\bar{\beta}}$ in equation (32), which determines the extent of time–variation in the skewness parameter, $\bar{\beta}_t$, is statistically significant for the majority of the series. For instance, regarding the four major crypto–currencies, $\alpha^{\bar{\beta}}$ is statistically different from zero for Ethereum, Litecoin and Ripple, and not for Bitcoin. In Figure 10 we report the filtered skewness parameter and volatilities for the

four major crypto–currencies.

We note that, except for Bitcoin, the conditional distribution is almost always positively skewed. Very interestingly, we find that when volatility increases the skewness parameter decreases, that is, an increment in the uncertainty for crypto–currencies time–series, generally decreases the likelihood of positive (excess) returns. This is particularly evident for Litecoin where, for example, during the increased volatility period exhibited at the beginning of the 2015, the skewness parameter decreases from 0.1 to -0.05, turning the conditional distribution to be negatively skewed for a short period. Results for the inclusion of time–varying kurtosis are very clear and indicate the lack of this feature for crypto–currencies financial time–series as reported in Panel (b) of Table 5.

6. Conclusion

In this paper, we analyzed the behaviour of crypto–currencies financial time–series of which Bitcoin is the most prominent example. The dynamic of those series is quite complex displaying extreme observations, asymmetries, and several nonlinear characteristics which are difficult to model. To accommodate those features, we developed a model able to incorporate: long memory, leverage effect, and time–varying higher order moments. The finite sample properties of the Maximum Likelihood estimator have been investigated via a Monte Carlo experiment. Results show that the MLE performs well in term of the bias and RMSE also in small samples. In the empirical application we apply this model to 289 crypto–currencies and we concentrate on the main four: Bitcoin, Ethereum, Ripple, and Litecoin. We found that a robust filter for the volatility of crypto–currencies time–series is strongly required by the data. Moreover, we find evidence of long memory in the volatility for some series, while for others a simpler specification is enough. We also find that, different from foreign exchange currencies, leverage effect has a substantial contribution in the volatility dynamic. On average, volatility increases more after negative shock than after positive shock as in the equity market; hence, crypto–currencies time–series incorporate the so–called leverage effect with some degree of heterogeneity across the series. We find evidence of time–varying skewness for some series and absence of time–varying kurtosis for the whole sample.

Implications of our results are in the area of risk management e.g. Value–at–Risk but also for investment purposes. To our knowledge, this is the first study investigating the statistical properties of a large set of crypto–currencies and indeed, there is much scope for future research in this new market.

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